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Capturing Correlation Effects on Photoionization Dynamics

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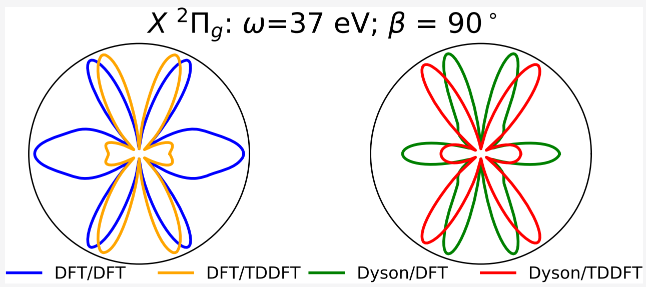

ABSTRACT: A highly correlated combination of the equation-of-motion coupled cluster (EOM-CC) Dyson orbital and the multicentric B-spline time-dependent density functional theory (TDDFT)-based approach is proposed and implemented within the single-channel approximation to describe molecular photoionization processes. The twofold objective of the approach is to capture interchannel coupling effects, missing in the B-spline DFT treatment, and to explore the response of Dyson orbitals to strong correlation effects and its influence on the photoionization observables. We validate our scheme by computing partial cross sections, branching ratios, asymmetry parameters, and molecular frame photoelectron angular distributions of simple molecules. Finally, the method has been applied to the study of photoelectron spectra of the $Ni(C_3H_5)_2$ molecule, where giant correlation effects completely destroy the Koopmans picture.

1. INTRODUCTION

The theoretical modeling of photoionization processes is a challenging field of quantum chemistry. It is the process of ejection of an electron to the continuum, and thus, acts as a fingerprint of the target molecule. The description of such processes requires treatment of both the bound target molecule and the outgoing electron at an equal footing. Even though there exist several accurate and robust methodologies for describing the bound state of the system, the same is not true for continuum electrons. In the case of the latter, the complication arises due to the oscillating behavior of the photoelectron wave function and the loss of the usual simple variational principle. Additionally, the basis set should be rich enough to sufficiently emulate the spatial regions far away from the molecule, as the continuum wavefunctions do not decay exponentially at large distances.

An old and popular strategy is to avoid the explicit description of the continuum orbital using the Stieltjes imaging and the equivalent Padé approximant formalisms.[1−10] These methods are based on the reconstruction of the cross sections from purely bound pseudo spectra. However, it is not possible to obtain individual channel cross sections and asymmetry parameters due to the lack of proper asymptotic boundary conditions, and the energy resolution is relatively modest. Still, these methods have proven quite successful when used in conjunction with highly correlated electronic structure methods such as linear response and equation-of-motion coupled cluster, and algebraic diagrammatic construction.[5−8,11,12]

On the other hand, various approaches have been developed to account for the true nature of the continuum electrons. A rather simple approximation is to describe the outgoing electron by analytical plane waves and Coulomb waves, or their orthogonalized variants.[13−15] However, these are shown to be not very accurate for studying a wide range of photon energies and specifically fail to reproduce spectral signatures associated with structured continua, like shape resonances and Cooper minima.[14,16] Alternatively, the continuum orbitals can be determined by stationary (variational) conditions, which can be formulated in technically different, but practically equivalent approaches like the R-matrix,[17,18] Schwinger variational,[19,20] complex Kohn,[21] least-squares or Galerkin methods.[22,23]

A major obstacle in the precise interpretation of photoelectron spectroscopic signals is the need to properly account for correlation effects. These can be separated into two broad categories. The first involves correlation within the bound states, initial and final (sometimes further divided into initial state configuration interaction (ISCI) and final state configuration interaction (FSCI)) and is taken care of using highly correlated wavefunction-based methods, like configuration interaction (CI),[24−26] perturbation theory,[25] coupled cluster,[13−16] and algebraic diagrammatic construction.[27] The second category involves interaction among different channels, also called interchannel coupling (IC). It can be widely divided into interactions between the electronic continua, the pure open-channel states, and interaction between an electronic continuum and a discrete excitation or a closed channel, which gives rise to autoionization resonances, and describes also the correlation between the bound electrons and the continuum, although it is

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difficult to draw unambiguous boundaries. In general, the interchannel coupling can be considered as the configuration interaction in the continuum (also known as the final continuum state configuration interaction, FCSCI) and is most complicated to capture, as it requires a general "close-coupling" treatment (see below).[28] The bound-state correlation, on the other hand, can be well described by quantum chemistry approaches, and then, rather straightforwardly included via Dyson orbitals[29,30] coupled to an accurate single-particle continuum, according to a single-channel formalism.[13−16,31,32] This is generally a good approximation, except for the near-threshold region, and far from autoionization resonances. Nevertheless, some interchannel coupling effects are sometimes important, especially when a weak channel is coupled to a much stronger one, with some intensity transfer taking place. At the lowest order, interchannel coupling is often rather well accounted for by the time-dependent density functional (TDDFT) approach, that is, the density functional theory analog of *ab initio* random-phase approximation (RPA), which includes coupling between singly excited configurations.

In this paper, we present a hybrid approach, named Dyson/TDDFT, coupling an accurate Dyson treatment of the bound-state correlation with the TDDFT continuum, generalizing and improving upon the Dyson-DFT scheme previously presented.[16] The results are reported for the photoionization observables, namely, partial cross sections (σ), branching ratios, asymmetry parameters (β), and molecular frame photoelectron angular distributions (MFPADs).

Preliminary results for the argon atom were presented in ref 16. Here, we will further validate the formalism by studying the outer-valence ionization dynamics of simple closed-shell molecules (H_2S, CS_2, C_3O_2, and C_3S_2). With reference to these molecules, we establish a link between the spectral signature and the nature of the bound orbital undergoing ionization. Furthermore, we attempt at identifying the particular scenarios where incorporation of correlation plays a key role. Finally, we examine the photoelectron spectrum of the controversial $Ni(C_3H_5)_2$ complex, the interpretation of which is not yet settled. It is a famous system where giant correlation/relaxation effects are active, highlighting the dual effects of bound-state correlation and interchannel coupling, thereby justifying the widespread applicability and accuracy of our scheme. An additional comparison with pure DFT/TDDFT results highlights the very different behavior of DFT and *ab initio* approaches.

2. THEORETICAL APPROACH

2.1. Overview. The theoretically relevant quantity for describing photoelectron spectroscopic observables is the photoelectron matrix element

$$\vec{D}^{if} = \langle \Psi_i^N | \vec{\mu} | \Phi_f \rangle \tag{1}$$

connecting the initial N-electron state (Ψ_i^N) and the final composite system described by the wavefunction Φ_f; $\vec{\mu}$ is the electric dipole moment. In principle, the final continuum can be described by the so-called "close coupling" form, which is equivalent to the configuration interaction (CI) approach in the continuum for the final composite system

$$\Phi_f \equiv \Phi_{EI\alpha} = \mathcal{A} \sum_J (\Psi_J^{N-1} \phi_{EJI\alpha}) + \sum_K \Phi_K^N C_{EKI} \tag{2}$$

The first term of eq 2 comprises all possible antisymmetrized products of $(N-1)$-electron final bound states, Ψ_J^{N-1}, accessible at energy E, coupled to continuum wavefunctions, $\phi_{EJI\alpha}$. The index J counts the different ionic (target) states included in the expansion, which give rise to an equivalent number of final independent solutions, indexed by I; α enumerates the independent continuum partial waves. The second term is a linear combination of N-electron localized basis functions, Φ_K^N, with coefficients C_{EKI}, accounting for both autoionizing states and short-range correlation effects. The *ab initio* computation using the complete expression is computationally very demanding and is usually restricted to small systems and low energies.[33−36] Several approximate formulations have been developed, differing both in the level of truncation of the close-coupling expression and in the treatment of the continuum orbitals $\phi_{EJI\alpha}$.[20,37−40]

We consider the single-channel approximation, where the second term of eq 2 is neglected and only one bound ionized state is considered at a particular energy. Hence, the expression for the final state becomes

$$\Phi_{EI\alpha} = \mathcal{A}(\Psi_I^{N-1} \phi_{EI\alpha}) \tag{3}$$

The simplified expression, eq 3, lacks coupling between open channels and correlation between the bound state and the continuum. However, these effects are weak, far from the ionization threshold and in the absence of autoionizing resonances. Interestingly, within the single-channel approximation, the many-particle photoelectron matrix element reduces to single-particle matrix elements[16,24,41]

$$D_\gamma^{if} = \langle \Psi_i^N | \mu_\gamma | \Phi_f \rangle = \langle \phi_{if}^d | \mu_\gamma | \phi_{EI\alpha} \rangle + \langle \eta_{if,\gamma}^d | \phi_{EI\alpha} \rangle \tag{4}$$

$$\phi_{if}^d = \langle \Psi_I^{N-1} | \hat{\psi}(x) | \Psi_i^N \rangle, \qquad \eta_{if,\gamma}^d = \langle \Psi_I^{N-1} | \mu_\gamma \hat{\psi}(x) | \Psi_i^N \rangle \tag{5}$$

where index γ labels a specific Cartesian component of the dipole operator, and ϕ_{if}^d and $\eta_{if,\gamma}^d$ are the Dyson orbital and the conjugate Dyson orbital, respectively. Note the use, in eq 5, of the field annihilation operator $\hat{\psi}(x)$, which can be expanded in terms of a complete basis as

$$\hat{\psi}(x) = \sum_q \phi_q(x) a_q$$

where a_q is the annihilation operator. These Dyson orbitals can be expressed entirely in terms of the bound-state wavefunction and embody all correlations necessary for their description.

The first Dyson term in eq 4 arises from a dipole transition of initial orbitals to the continuum, multiplied by the overlap of the passive ones. The second term, conjugate, arises from a dipole transition within the bound orbitals, and an overlap of the remaining one with the continuum.[16,24,41] It appears because of nonorthogonality of the continuum to the initial bound eigenstate and is zero if all bound orbitals in the initial state are orthogonal to it. Moreover, its relative intensity with respect to the Dyson term decreases with increasing electron energy. Apart from special cases, its contribution is generally very small and is neglected. A well-established formalism to obtain the photoelectron matrix element uses single determinants built from the occupied orbitals at the ground state, and variationally computes the continuum. This is known as the frozen Hartree−Fock, or static exchange (SE) method. Alternatively, an equivalent formulation in the DFT regime, namely the static exchange DFT (SE-DFT) is obtained employing both bound

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and continuum orbitals as eigenstates of the Kohn−Sham DFT Hamiltonian, determined from the ground-state density.[42,43] At this level, employing a common set of orthonormal orbitals (bound + continuum), the continuum is orthogonal to all initial orbitals and the overlapping term is zero. The same is true at the SCI or TDDFT levels. When correlation is included in the bound states, or relaxation is allowed, orthogonality breaks down. In the case of very strong orbital relaxation, like in the case of double-core holes, the conjugate term may actually become comparable with the direct one.[44] When employing a correlated ionic state, the SE-DFT continuum is no longer variational, although combining it with the Dyson orbital is still a good approximation, as previously shown.[16,31,32,45] We refer to it as Dyson/DFT formulation.

The simplest method that includes interchannel coupling effects (correlation between the continuum channels), while still treating the bound states as a single determinant, is the singly excited CI (CIS) approach or the Tamm−Dancoff approximation (TDA). An additional correlation may be included via the random-phase approximation (RPA), in the linear response formalism, still conserving the essential CIS structure of excitation. Similarly, the equivalent formulation within DFT is the TDDFT approach. It is interesting that TDDFT (or RPA) can be seen in two equivalent, but physically quite distinct, ways. The one most familiar to chemists is a basis set formulation, known as the Casida approach,[46] which clearly shows the correspondence to CIS. The second is from the response viewpoint, which describes the screening of the external exciting field due to the response of the electron cloud.[47−49] The fundamental equations governing the linear response are

$$\delta\rho = \chi V^{\mathrm{SCF}}, \qquad \delta V = K\delta\rho, \qquad V^{\mathrm{SCF}} = \mu^{\mathrm{ext}} + \delta V \tag{6}$$

where $\delta\rho$ is the first-order change in the electron density induced by the field, χ is the electric susceptibility, δV is the first-order change in the potential induced by a density change $\delta\rho$, and K is the linear kernel relating the two quantities. The final full potential (V^{SCF}) is the sum of the external perturbing potential (μ^{ext}) and the response potential (δV). It is known that the effect—i.e., interchannel coupling in the first interpretation, and screening in the second—may be quite strong, especially close to the threshold. While in the first viewpoint, the coupling of this continuum description with a single-channel formulation may appear strange, it is more natural in the latter. In the response framework, the effect of electron polarization gives rise to an effective potential, which is the sum of the external dipole field and the electronic response potential. The evaluation of the photoelectron matrix element then boils down to replacing the pure dipole field in eq 4 by the effective potential V^{SCF}

$$D_\gamma^{if} = \langle \Psi_i^N | V_\gamma^{\mathrm{SCF}} | \Phi_f \rangle \tag{7}$$

From this expression, introducing a Dyson/TDDFT approach comes naturally, yielding

$$D_\gamma^{if} = \langle \phi_{if}^d | V_\gamma^{\mathrm{SCF}} | \phi_{EI\alpha} \rangle \tag{8}$$

We have implemented and discussed the Dyson/TDDFT formulation, eq 8, in this article. The length gauge has been used throughout. The details of the algorithm are described in the following subsections.

2.2. Linear Combination of Atomic Orbital (LCAO) B-Spline TDDFT Continuum. Solving for the continuum orbital within the TDDFT approach is quite involved. As the TDDFT formalism can be considered an extension to the Kohn−Sham DFT (KS-DFT), we start from the latter as input for computing the TDDFT continuum orbitals. At the KS-DFT stage, the continuum orbitals ($\phi_{\epsilon lm}$) are computed by solving the Schrödinger equation in the angular momentum representation, with l and m as usual angular momentum quantum numbers

$$H_{\mathrm{KS}}\phi_{\epsilon lm} = E\phi_{\epsilon lm} \tag{9}$$

where

$$H_{\mathrm{KS}} = -\frac{1}{2}\Delta - \sum_N \frac{Z_N}{|\bar{r} - \bar{R}_N|} + \int \frac{\rho(\bar{r}')\mathrm{d}\bar{r}'}{|\bar{r} - \bar{r}'|} + V_{\mathrm{XC}}[\rho(\bar{r})] \tag{10}$$

Here, H_{KS} is the KS Hamiltonian and $\rho(\bar{r})$ is the ground-state electron density. For V_{XC}, we have here chosen the LB94 exchange−correlation potential due to its proper description of the asymptotic Coulomb behavior.[43] The continuum KS wavefunctions are then obtained as eigenvectors with minimum modulus eigenvalue of the energy-dependent matrix $\mathbf{A}^\dagger \mathbf{A}$

$$\mathbf{A}^\dagger\mathbf{A}(E)c = ac, \qquad \mathbf{A}(E) = \mathbf{H} - E\mathbf{S} \tag{11}$$

where $\mathbf{H}$ and $\mathbf{S}$ are the Hamiltonian and overlap matrices, respectively, E is photoelectron kinetic energy, c is the eigenvector, and a is the minimum modulus eigenvalue.[50]

From eqs 6, one can obtain an equation for V^{SCF}

$$\begin{aligned} V^{\mathrm{SCF}} &= \mu^{\mathrm{ext}} + K\delta\rho \\ &= \mu^{\mathrm{ext}} + K\chi V^{\mathrm{SCF}} \end{aligned} \tag{12}$$

We then solve directly for the response potential (V^{SCF}) as the prime dynamical variable, using a noniterative numerical algorithm.[50] The kernel K is the sum of the Coulomb potential and the linearized exchange−correlation response

$$K(\bar{r}, \bar{r}') = \frac{1}{|\bar{r} - \bar{r}'|} + \delta(\bar{r} - \bar{r}')\frac{\delta V_{\mathrm{XC}}}{\delta\rho} \tag{13}$$

By representing V^{SCF} as well as the operators K and χ in the B-spline basis, the integrodifferential equation, eq 12, is converted into a linear algebraic one

$$(\mathbf{K}\boldsymbol{\chi} - \mathbf{1})\mathbf{V}^{\mathrm{SCF}} = -\boldsymbol{\mu}^{\mathrm{ext}} \tag{14}$$

The computationally expensive part is the calculation of the KS (noninteracting) linear susceptibility, which is energy-dependent and is computed for each photon energy by solving the inhomogeneous first-order perturbative equations. The completeness of the basis ensures convergent solutions, and the full details are available in ref 50.

As anticipated, we employ a basis set expansion, with primitives constructed as products of radial B-splines ($B_j(\bar{r})$) and real spherical harmonics ($Y_{lm}(\theta, \phi)$) located on different centers, in the spirit of a linear combination of atomic orbitals (LCAO)

$$\chi_{jlm}(\bar{r}, \theta, \phi) = \frac{1}{r}B_j(\bar{r})Y_{lm}(\theta, \phi) \tag{15}$$

The B-splines are defined on a given interval, say $[0, R_{\max}]$, divided into subintervals by a nondecreasing sequence of points (knots). Each B-spline is made of different polynomial pieces over n successive knots (n is the fixed order of the polynomials) joined with a high degree of continuity between adjacent subintervals, and is zero outside, so that exactly n splines overlap on each subinterval, and linear independence is optimally

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assured. The B-splines can approximate any smooth function in the given interval with arbitrary accuracy by decreasing the knot spacing and increasing the polynomial order.[51] They enjoy the flexibility of finite difference methods on a grid, with the ease of manipulation of basis sets via linear algebra algorithms. In our implementation, a long-range expansion is located on the origin of the molecule (one center, OCE), supplemented with short expansions around the nuclei, to account for Coulomb cusps. The OCE expansion defines a sphere large enough that bound states are negligible outside, and the oscillating continuum has reached the asymptotic limit and can be fitted to the known analytic solutions (coulomb functions for photoionization).[42]

2.3. EOM-CCSD Dyson Orbitals. The one-electron Dyson orbital is defined as the overlap between the bound initial N-electron and final $(N-1)$-electron states, given by

$$\phi_{if}^{d} = \sqrt{N}\int \Psi_i^N(x_1, x_2, ..., x_N)\Psi_f^{N-1}(x_2, ..., x_N)\mathrm{d}x_2 \ldots \mathrm{d}x_N \equiv \sum_p \gamma_p \phi_p \tag{16}$$

It can be further reduced to a linear combination of ground-state molecular spin−orbitals (ϕ_p), with combination coefficients γ_p. In a single determinant description, considering ionization from the kth molecular orbital, the Dyson orbital reduces to the initial occupied orbital ϕ_k, and the corresponding ionization energy is equal to the negative kth orbital energy, according to Koopmans' theorem[52]

$$\phi_{if}^{d} = \phi_k \text{ and } \gamma_p = \delta_{kp} \tag{17}$$

Here, we use the EOM-CCSD framework,[53,54] where the bound states are obtained by diagonalizing the matrix representation of the non-Hermitian similarity-transformed Hamiltonian, $\bar{H} = \exp(-T)H\exp(T)$, on the reference and the truncated excitation space

$$\bar{\mathbf{H}}\mathbf{R}_m = E_m\mathbf{R}_m, \qquad \mathbf{L}_m^\dagger\bar{\mathbf{H}} = \mathbf{L}_m^\dagger E_m \text{ and } \mathbf{L}_m^\dagger\mathbf{R}_n = \delta_{mn} \tag{18}$$

In connection to ionization processes, the final bound $(N-1)$ state wavefunction can be expressed (for CCSD) as

$$|\Psi^{N-1}\rangle = R^{\mathrm{IE}}|\Psi_{\mathrm{CCSD}}^N\rangle = R^{\mathrm{IE}}\exp(T_1 + T_2)|\Phi_0\rangle \tag{19}$$

where

$$R^{\mathrm{IE}} = \sum_i r_i a_i + \frac{1}{2}\sum_{aij} r_{ij}^a a_a^\dagger a_j a_i \tag{20}$$

and $|\Psi_{\mathrm{CCSD}}^N\rangle$ and $|\Phi_0\rangle$ are the CCSD ground-state wavefunction and the reference Hartree−Fock Slater determinant for the N electron state, respectively. T, R^{IE}, and $(L^{\mathrm{IE}})^\dagger$ are excitation operators from a reference determinant. The operator R^{IE} (and similarly L^{IE}) is not electron-conserving, as shown in eq 20. Indices i and j refer to occupied orbitals and the index a to virtual orbitals. In coupled cluster theory, bra (left) and ket (right) states are not each other's adjoint.[55] As a consequence, the left ($\phi^{d,\mathrm{L}}$) and right ($\phi^{d,\mathrm{R}}$) EOM-CCSD Dyson orbitals are different, i.e., their coefficients $\gamma_p^{\mathrm{L}}/\gamma_p^{\mathrm{R}}$ are distinct. For ionization from the ground state, the γ_p coefficients are[15]

$$\gamma_p^{\mathrm{L}} = \langle\Psi_{\mathrm{L}}^{N-1}|a_p|\Psi_{\mathrm{R}}^N\rangle = \langle \mathrm{HF}|L^{\mathrm{IE}}\exp(-T_2 - T_1)a_p\exp(T_1 + T_2)|\mathrm{HF}\rangle \tag{21}$$

and

$$\gamma_p^{\mathrm{R}} = \langle\Psi_{\mathrm{L}}^{N}|a_p^\dagger|\Psi_{\mathrm{R}}^{N-1}\rangle = \langle \mathrm{HF}|(1+\hat{\Lambda})\exp(-T_2 - T_1)a_p^\dagger R^{\mathrm{IE}}\exp(T_1 + T_2)|\mathrm{HF}\rangle \tag{22}$$

A suitable inclusion of the bound-state correlations and even treatment of nonsingle determinant states (for example, satellites) as complex open shells can be obtained by employing the Dyson orbitals.

The squared norm of the Dyson orbital, defined as

$$R_F = \|\phi^d\|^2 = \sum_p \gamma_p^2 \tag{23}$$

is the spectral strength, also called the pole strength or spectroscopic factor, of the final state Ψ^{N-1}. The values of R_F are used as the intensity of bands when simulating the photoelectron spectrum. As a known effect of inclusion of correlation, the spectral strengths of the primary ionic states are lowered and, at the same time, additional states, i.e., satellite or shake-up states, which are characterized by further electronic excitations, gain intensity. As the right and left Dyson orbitals are different in EOM-CC, one may ponder how to compute the spectroscopic factor, for instance, whether one should take the square norm of just one of the two Dyson orbitals (left or right), or to use the product of their norms (geometric norm), or a third recipe altogether.[56,57] Here, we propose to use, as a spectroscopic factor, the dot product of the corresponding left and right Dyson orbital pairs

$$R_F = \sum_p (\gamma_p^{d,\mathrm{L}}\gamma_p^{d,\mathrm{R}}) \tag{24}$$

It is worth noticing, nonetheless, that the differences between the numerical results of the different definitions are almost negligible (see, e.g., Figure S2).

2.4. Photoionization Parameters. The differential partial cross section in atomic units (in the molecular frame) is given by

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\vec{k}} = 4\pi^2\alpha\omega|D^{if}|^2 \tag{25}$$

where α is the fine structure constant, ω is the photon energy, and $\vec{k}$ is the momentum of the photoelectron in the molecular frame.

In the case of randomly oriented molecules, following the standard manipulations as adopted by Chandra,[58] one obtains

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\vec{k}} = \pi\alpha\omega(-1)^{m_r}\sum_L A_L(k)P_L(\cos\theta) \tag{26}$$

where m_r specifies the polarization of light with possible values of 0, +1, and −1 for linearly polarized (LP), left circularly polarized (LCP), and right circularly polarized (RCP) light, respectively; θ is the scattering angle between $\vec{k}$ and the laboratory frame defined by the electric field (for LP) or direction of propagation of light (for CP). The coefficients $A_L(k)$ are given by

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$$A_L(k) = (2L+1)\begin{pmatrix} 1 & 1 & L \\ m_r & -m_r & 0 \end{pmatrix} \sum_{\substack{lm\gamma \\ l'm'\gamma'}} (-1)^{m+\gamma} \sqrt{(2l+1)(2l'+1)} \times \begin{pmatrix} l & l' & L \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & l' & L \\ -m & m' & (m-m') \end{pmatrix} \begin{pmatrix} 1 & 1 & L \\ \gamma' & -\gamma & (\gamma-\gamma') \end{pmatrix} \times \frac{1}{2}(D^{\mathrm{R}}_{\epsilon lm,\gamma} D^{\mathrm{L}*}_{\epsilon l'm',\gamma'} + D^{\mathrm{L}}_{\epsilon lm,\gamma} D^{\mathrm{R}*}_{\epsilon l'm',\gamma'}) \tag{27}$$

P_L are Legendre polynomials of order L, where $L = 0, 1, 2$, and $D^{\mathrm{R}}_{\epsilon lm,\gamma}$ and $D^{\mathrm{L}}_{\epsilon lm,\gamma}$ indicate specific right and left photoelectron dipole matrix elements in the Cartesian direction γ. For more details, refer to the Supporting Information of ref 16. Defining

$$\frac{\sigma}{4\pi} = \pi\alpha\omega(-1)^{m_r} A_0; \qquad D = \frac{A_1(m_r = 1)}{A_0}; \qquad \beta = \frac{A_2(m_r = 0)}{A_0} \tag{28}$$

where σ is the partial cross section, β is the asymmetry parameter, and D is the dichroic parameter, allows us to rewrite the final expression for the differential cross section as

$$\frac{d\sigma}{d\vec{k}} = \frac{\sigma}{4\pi}\left[1 + \left(-\frac{1}{2}\right)^{|m_r|} \beta P_2(\cos\theta) + m_r D P_1(\cos\theta)\right] \tag{29}$$

Equation 29 corroborates that the angular distribution of the photoelectrons only depends on the cross section σ, asymmetry parameter β, and the dichroic parameter D. We will in the following only consider linearly polarized light, i.e., $m_r = 0$, so eq 29 boils down to

$$\frac{d\sigma}{d\vec{k}} = \frac{\sigma}{4\pi}[1 + \beta P_2(\cos\theta)] \tag{30}$$

Besides partial cross-sections and asymmetry parameters, we will also study the branching ratio, defined as the ratio between a specific cross section and the sum of all considered cross sections. Additionally, molecular frame photoelectron angular distributions (MFPADs) are computed as the polar plot of the differential partial cross section, following the general treatment described in, e.g., refs 58, 59.

3. COMPUTATIONAL DETAILS

Experimental geometries from the NIST compilation[60] have been used for H_2S and CS_2. The geometry of the nickel complex is the same as used in ref 61. They are reported in the Supporting Information file as Cartesian coordinates. The ground-state electronic densities were obtained from the ADF program package[62] using the LB94 exchange−correlation potential with the DZP basis set.[63] Following this step, both the bound and continuum orbitals were then computed in the LCAO B-spline basis, with a suitable choice of parameters, which is given in Table 1. With this choice of parameters, we ensured accurate convergence of the photoionization results.

The ionization energies and corresponding Dyson orbitals were computed at the EOM-CCSD level using Q-Chem.[64] CCSDR(3) results for the ionization energies of CS_2 were obtained using Dalton. A comparison of the photoelectron spectra obtained employing different descriptions of the bound states is shown in Figure S1. Dunning's correlation consistent basis sets augmented with polarization and diffuse functions (aug-cc-pVXZ) were used for the small molecules (X = T for H_2S; X = D for CS_2, C_2O_3, and C_2S_3). In the case of the nickel bisallyl complex, the cc-pV(D + d)Z basis set for the Ni atom and the DZP basis for C and H were employed. To reduce the computational cost, we froze the 1s orbital of the six carbon atoms and up to the 2p orbitals of Ni. Note that we present the results as per conventional symmetry notations, even though in the Q-Chem calculations a non-Mulliken representation is adopted. All remaining calculations involving the continuum have been performed using the Tiresia code.[42]

Table 1. Parameters Used in the LCAO B-Spline Code[a]

molecule	no. of knots	L_{max}	R_{max}	LCAO descriptor l_{max}	LCAO descriptor r_{max}
H_2S	100	12	25.0	1(H)	0.600(H)
CS_2	100	20	25.0	2(S)	1.000(S)
C_3O_2	100	18	25.0	2(C,O)	0.992(C), 1.200(O)
C_3S_2	100	16	25.0	2(C,S)	1.000(C), 1.900(S)
$Ni(C_3H_5)_2$	100	16	25.0	2(C),1(H)	1.300(C), 0.700(H)

[a] L_{max} and l_{max} are the maximum angular momentum employed in the one-center expansion at the origin and on each off-center atom, respectively. R_{max} and r_{max} (both in au) are the maximum radial grid length from the origin and from the off-center atoms, respectively.

Throughout the article, we follow the conventional nomenclature of designating the first ionization band as *X*, the second as *A*, the third as *B*, and so on for the outer-valence ionizations. At higher ionization energies, satellite bands and breakdown of the orbital picture appear. We have not considered this region in detail, as very few accurate photoionization parameters are available, and the present correlation level is not generally accurate enough for a quantitative description.

Here, we first validate the Dyson/TDDFT approach over small molecules (H_2S, CS_2, C_3O_2, and C_3S_2) by comparing the performance of the Hartree−Fock molecular orbitals, the KS-DFT molecular orbitals, and the EOM-CCSD Dyson orbitals for the bound-state description, in conjunction with the corresponding LCAO B-spline DFT and TDDFT continuum. To avoid discussion of similar results and due to the lack of experimental profiles for C_3O_2 and C_3S_2, they are presented in the Supporting information. Second, we test our approach on the highly correlated spectrum of the $Ni(C_3H_5)_2$ complex, whose interpretation is still controversial.

As anticipated, results are reported for ionization energies, spectroscopic factors R_F branching ratios, partial cross sections σ, asymmetry parameters β, and MFPADs.

4. RESULTS AND DISCUSSION

4.1. Hydrogen Sulfide. The outer-valence photoionization spectrum of H_2S has been reported by several authors, and we here specifically refer to the experimental results by Baltzer et al.[65] Our results for the ionization energies and pole strengths are collected in Table 2. The photoelectron spectrum is reproduced in Figure S1. It comprises three well-separated bands due to the ionization from the $2b_1$, $5a_1$, and $2b_2$ molecular orbitals. All of the three (right) EOM-CCSD Dyson orbitals (shown as insets to Figure 1) have significant contributions from the central sulfur atom. The *X* band is purely due to ionization

Table 2. H_2S: Hartree−Fock Ionization Energies According to Koopmans' Theorem, LB94 Ground-State KS Eigenvalues, and EOM-CCSD Ionization Energies Versus the Experiment[a]

state	HF	LCAO KS	EOM-CCSD	R_F	exp.
$X\ ^2B_1$	10.48	10.85	10.42	0.921	10.46
$A\ ^2A_1$	13.66	13.72	13.43	0.913	13.45
$B\ ^2B_2$	16.05	15.57	15.69	0.914	15.55

[a]All values are in eV. The EOM-CCSD pole strengths are reported in the second last column. Experimental results are taken from ref 68.

from the $3p_x$ lone pair electron over the S atom, thereby having the largest atomic contribution among the three. The Dyson orbital, corresponding to the A band, is quite similar to the $5a_1$ molecular orbital. Both A and B bands are strongly bonding, with mainly S−H σ bond nature, with the B band showing a broader vibrational envelope. As the transitions occur from orbitals having some S 3p character, it is anticipated that their cross-sections and asymmetry parameters will show the characteristic behavior of the atomic component. The ionization from 3p orbitals of the third-period elements exhibits Cooper minimum. For sulfur, this leads to a pronounced dip in the asymmetry parameter at around 40 eV. Remarkably, the dip is the largest in the case of the X band, corroborating the fact that it has the maximum sulfur 3p contribution. In general, the orbital composition is reflected in the photoionization parameters through the dipole matrix elements with the continuum. In the molecular case, the loss of symmetry results in many partial waves contributing, so it becomes difficult to analyze contributions in detail. However, typical atomic features survive in the molecular cross section. They are very useful to identify atomic contributions in the initial orbitals, and even quantify trends, like the participation of a specific AO to different molecular orbitals, or to analogous orbitals in different compounds. The depth of a Cooper minimum is one example, as also seen in the following CS_2 case; other examples are delayed onset characteristics of d and f AOs, especially in the heavier atoms, atomic shape resonances like d → f excitations, and characteristic autoionization resonances. Other features are characteristics of bonds, like shape resonances that can be associated with excitations into σ^* antibonding orbitals, or diffraction features, which are very sensitive to bond length and molecular geometry.[66,67]

The branching ratio profiles show some features near the threshold and then become flat above 60 eV. As the Dyson orbitals are very similar to the corresponding Hartree−Fock and DFT molecular orbitals, no noticeable difference is observed upon variation in the bound-state description. On the other hand, the spectral feature is sensitive to the use of either DFT or TDDFT continuum orbitals. A maximum appears in all branching ratio profiles of the B band using the TDDFT continuum at around 33−35 eV, which is not captured by the DFT approach. Insets zooming into this region are shown in Figure 1. Also, the discrepancy is clearly manifested in the asymmetry parameter profiles. The DFT continuum consistently underestimates the photon energy corresponding to the Cooper minimum by 6−8 eV.

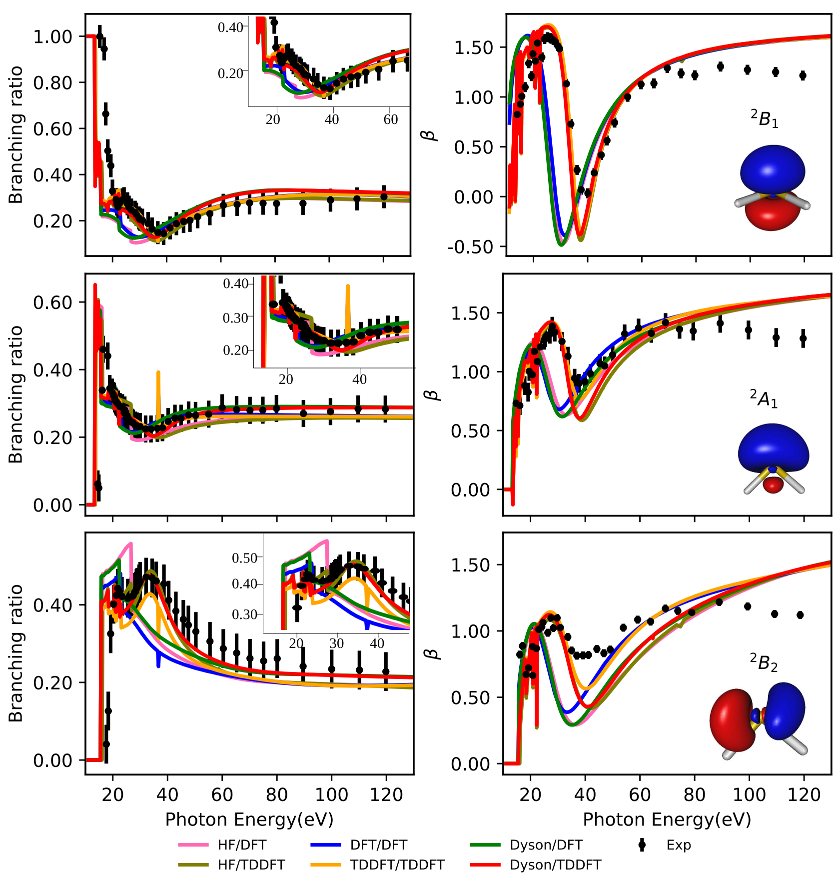

Figure 1. H_2S: Branching ratio and asymmetry parameter for the three outer-valence ionizations. Experimental results are taken from ref 65. Zoomed-in branching ratio and right EOM-CCSD Dyson orbitals are plotted as insets.

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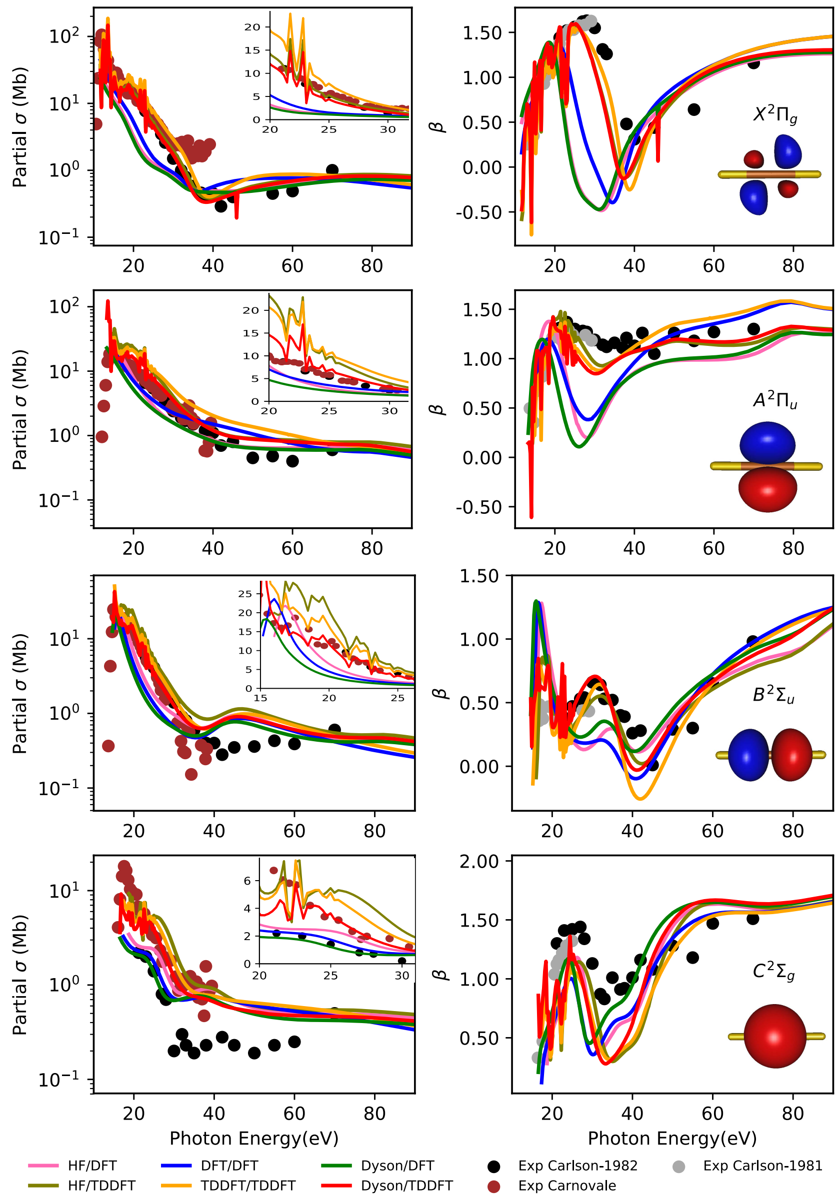

Figure 2. CS_2: partial cross section and the asymmetry parameter for the four lowest energy ionizations. Experimental results are taken from refs 69−71. Zoomed-in partial cross sections and the right EOM-CCSD Dyson orbitals are shown as insets.

4.2. Carbon Disulfide. For CS_2, we studied the five lowest energy ionizations. The four outer-valence bands are quite straightforward, in the sense that they are solely characterized by ionization from a particular molecular orbital, with R_F greater than 0.8. The partial cross sections and asymmetry parameters are plotted as functions of photon energy in Figure 2. Table 3 summarizes our results for the ionization energies (and the EOM-CCSD pole strengths) versus the experiment. We refer to Figures S1 and S3 for renderings of the photoelectron spectrum.

In previous work,[69] it was suggested that the strong enhancement of the cross section close to the threshold is due to the presence of a shape resonance. This suggestion is supported by the present approach, as a strong enhancement appears both in the DFT and TDDFT results. Curiously, however, the effect of TDDFT is significantly stronger in the highest occupied molecular orbital (HOMO) channel, compared to the following valence ionizations. This indicates that the dipole transition moments to the continuum from

Table 3. CS_2: Hartree−Fock Ionization Energies According to Koopmans' Theorem, LB94 Ground-State KS Eigenvalues, EOM-CCSD and CCSDR(3) Ionization Energies Versus the Experiment[a]

state	HF	LCAO KS	EOM-CCSD	R_F	exp.
$X\ ^2\Pi_g$	10.20	11.38	9.91 (9.85)	0.897	10.1
$A\ ^2\Pi_u$	14.37	14.27	13.20 (12.87)	0.825	12.9
$B\ ^2\Sigma_u$	15.78	15.07	14.45 (14.25)	0.873	14.6
$C\ ^2\Sigma_g$	18.50	17.15	16.42 (16.07)	0.814	16.2
$D\ ^2\Pi_u$			17.78 (14.60)	0.014	17.2
$^2\Sigma_u$	27.95	23.41	22.78 (21.46)	0.730	
$^2\Pi_u$			21.67 (17.29)	0.058	

[a]The perturbatively corrected ionization energies, obtained using the CCSDR(3) method, are shown in parentheses. All energy values are in eV. The EOM-CCSD pole strengths are reported in the second last column. The experimental results are taken from ref 69.

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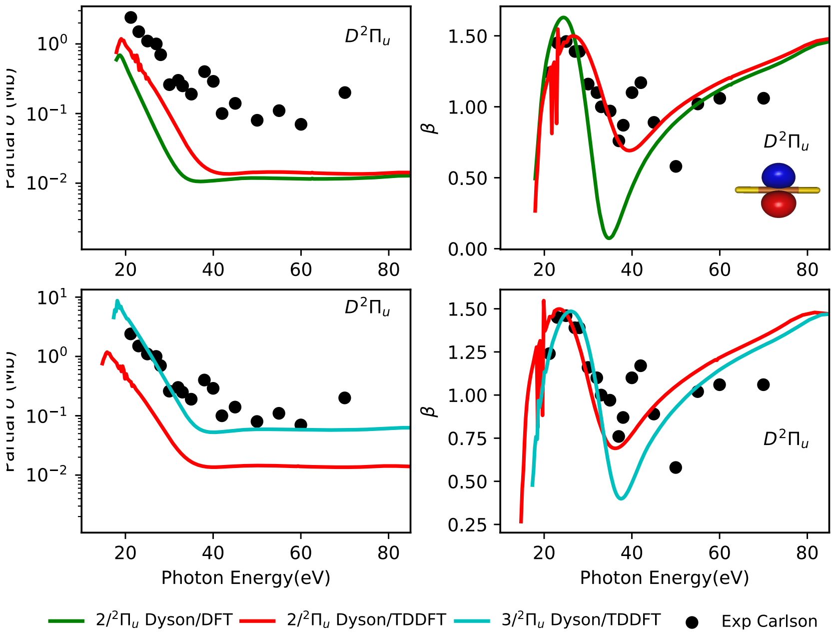

Figure 3. CS_2: partial cross section and the asymmetry parameter of satellite states using EOM-CCSD Dyson orbitals. EOM-CCSD and CCSDR(3) ionization energies have been used in the top and bottom panels, respectively. Experimental results are taken from ref 69. The right Dyson orbital is shown as an inset.

different initial orbitals are differently affected by interchannel coupling. Starting from the lowest energy ionization, i.e., from the HOMO $(2\pi_g)^{-1}$, there is a remarkable difference between the partial cross sections computed using DFT and TDDFT continua in the near-threshold region (below 40 eV). The DFT spectrum underestimates the intensity in the region between 15 and 35 eV, with respect to the experiment. The TDDFT continuum, on the other hand, gives a substantial improvement. Moreover, several sharp structures are obtained close to the threshold due to autoionization resonances. These are generally washed out by vibrational motion in the experiment. Experimentally, these sharp peaks in the *X* band, near-threshold, have been attributed to auto-ionizing Rydberg levels converging to higher-lying ionic states in addition to direct single-orbital ionizations.[71] Then, we see that the TDDFT description of the continuum incorporates a new phenomenology, which is lacking in the DFT approach. However, both approaches are capable of capturing the Cooper minima. In the case of TDDFT, the agreement with the experiment is excellent. The asymmetry parameter for the *X* band shows the typical nature as seen for atomic S 3p orbital.

Next, for the *A* band, all combinations of methods perform equally well in terms of matching with the experimental cross section. Astonishingly, all of the methods fail to capture the intensity increase in the near-threshold region. However, the accuracy of the various method differs markedly when looking at the asymmetry parameter. The Dyson/TDDFT approach reproduces the tail region of the spectrum above 50 eV the best. The DFT continuum, on the other hand, overestimates the dip at the Cooper minima by about 1.0 units.

The third band arises dominantly due to electron ejection from the $5\sigma_u$ orbital. The DFT continuum approach gives a rapidly decreasing cross-section profile. Even though the TDDFT continuum shows minimal deviation from the experimental profile up to 40 eV, it still lacks the narrow threshold feature. The importance of correlation in the bound-state description using Dyson orbitals is again demonstrated in the asymmetry parameter profile. The Dyson/TDDFT approach gives the closest representation of the experiment. The pure TDDFT computation overestimates the broad dip in the minima at 40 eV, whereas DFT continuum computations diminish the extent of increase at the maxima. Last, it is difficult to comment on the performance of the combination of methods in comparison to the experiment for band *C*, since the two experiments do not match with one another.

The outermost $(2\pi_g)$ orbital is a pure combination of sulfur 3p orbitals in a minimal basis set description, the following $2\pi_u$ comprises some C 2p, and even the next σ_u is basically built from the same atomic orbitals, still dominated by sulfur 3p. Still, the changes are clearly reflected in the cross section and more strongly in the β parameter, and the variations are well captured by the theoretical approach.

According to the single-particle ionization pathway available from considering Koopmans' theorem, the fifth band should be due to ionization from the $4\sigma_u$ orbital. However, the ionization energy for $(4\sigma_u)^{-1}$ lies much higher in energy than the experimental band (27.95 eV in Table 3). At the EOM-CCSD level, the corresponding $^2\Sigma_u$ state, with a significant amount of primary ionization, is found at 22.78 eV. In ref 69, it was suggested that the fifth band is due to the main satellite peak of π_u character. Using EOM-CCSD, we find the fifth band of Π_u character at 17.78 eV, reasonably close to the experimental value of 17.2 eV. However, this ionization has a very low Dyson pole strength (0.014) and is predominantly of double-excitation character. An additional Π_u ionization with a sizeable amount of primary single-excitation character is found at 21.67 eV, with a pole strength of 0.058. Since states of double-excitation character are not so well described by EOM-CCSD (their energy is in general overestimated), we tried and applied a perturbative correction to the ionization energies using the CCSDR(3) method.[72] The results are also reported in Table 3.

In the two top panels of Figure 3, we present the partial cross section and asymmetry parameter for the first of the two EOM-CCSD Π_u satellites and the two different continuum

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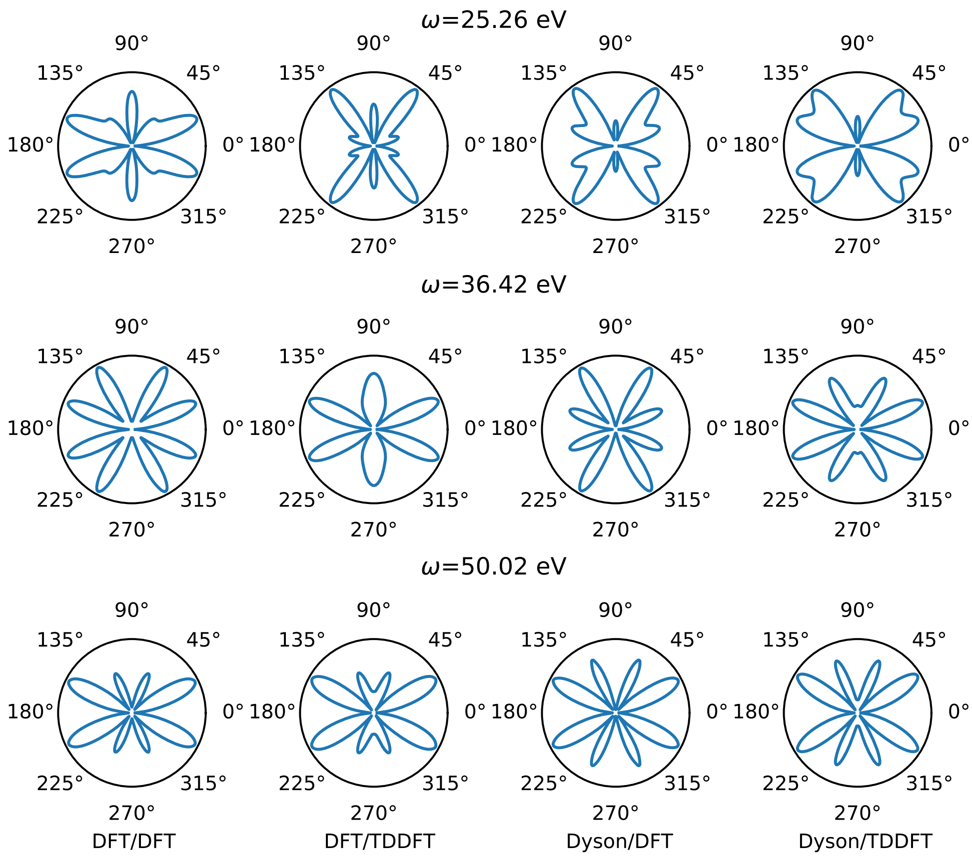

Figure 4. CS_2: MFPADs of the $X\ ^2\Pi_g$ ionized state. The polarization vector is parallel to the molecular axis. All plots have been normalized to a common value for the maximum.

descriptions. The two panels at the bottom report the same data using the CCSDR(3) ionization energies and CCSD Dyson pole strengths for both Π_u satellites and the TDDFT continuum. We refer to Figure S4 for the corresponding plots using the second Σ_u ion.

If we describe the *D* band using the data for the first EOM-CCSD Π_u satellite, the partial cross section is underestimated by one order of magnitude owing to the significantly small R_F value. This is considerably improved when we use the EOM Dyson orbital of the second Π_u satellite. The asymmetry parameter is quite accurately reproduced using the Dyson orbitals of the Π_u symmetry satellites, contrary to what is obtained with the higher-energy Σ_u one. This corroborates the analysis in the experimental study.[69] It is also quite evident from Figure 3 that the TDDFT continuum orbitals better represent the actual scenario than the DFT continuum orbitals.

Finally, as a signature of the final states, we have computed the molecular frame angular distributions, as shown in Figures 4 and S5−S10. Our choice of photon energies ω was driven by the desire to investigate the change in the angular distribution near the Cooper minimum at around 37 eV. In the case of the parallel orientation for the *X* band, the emission profiles obtained from the DFT and the TDDFT treatments of the continuum are quite different, although maintaining a butterfly-like shape. Due to the symmetry of the molecule, the forward and backward emissions are mirror images of one another.

4.3. Nickel Bisallyl Complex. We now investigate the giant correlation effects necessary for the description of transition metal compounds. The $Ni(C_3H_5)_2$ complex, see Figure 5, is an interesting molecule due to the strong disagreement between

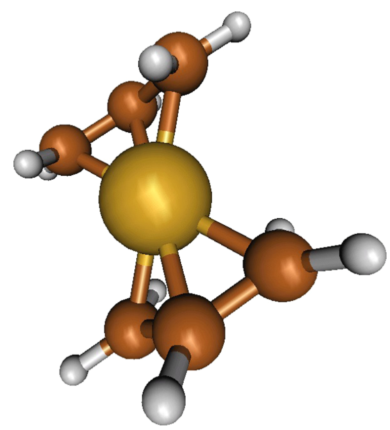

Figure 5. Nickel−bisallyl complex.

experimental evidence and results from Hartree−Fock molecular orbital calculations. The molecule is a classic example of the complete breakdown of Koopmans' theorem. It exhibits large relaxation effects, especially in the case of ionization from the Ni d orbitals. The d orbitals are embedded deep within the valence shell at the Hartree−Fock level, but become the lowest bound at the ΔSCF level, which provides relaxation energies up to 10 eV for such ionizations.[73] This is partly exaggerated because of the lack of correlation, which alleviates the strong interelectronic repulsion of the compact d shell. Despite the intense effort in the past, the assignment of the spectrum is still not firmly settled. It is generally accepted that the first three bands comprise both four metal 3d (^2A_g and ^2B_g) ionizations, and the outermost π allyl

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ionization ($7a_u$), but the specific ordering is debated. A careful experimental study, based on trends among substituted allyl compounds,[74] reinforced by later synchrotron radiation experiments,[75,76] favors metal d orbital ionization as the outermost, while the most elaborate theoretical calculations give allyl $7a_u$ as the least bound, but again details vary considerably.

In Table 4, we report a comparison between the experiment and computed ionization energies obtained from Koopmans' theorem at the Hartree−Fock level and from EOM-IP-CCSD/cc-pV(D+d)Z(Ni)/DZP(C,H) calculations. This choice of basis set is a compromise between accuracy and computational cost. We note, however, that using the aug-cc-pVDZ basis set on all atoms, the photoelectron spectrum does not vary significantly; see Figure S13 for more details. We shall not discuss the assignment in detail; however, note that while we agree in indicating $7a_u$ as the outermost ionization, it is almost degenerate with the following metal 3d bands, and this is probably the reason for the difficulty to ascertain the correct ordering. Looking at the calculated spectrum (Figure 6), we observe that although the outermost structure is not perfect—with bands 2 and 3 that should be closer to band 1 and too large a gap between bands 4 and 5—it is overall quite reasonable. We also obtain a tiny hump region between bands 9 and 10, in the range of photon energies from 16.9 to 17.8 eV, arising due to transitions having very low Dyson square norm. These target-bound ionization states are also an outcome of correlation included in the EOM-CCSD Dyson orbital framework.

Table 4. $Ni(C_3H_5)_2$: Hartree−Fock Ionization Energies According to Koopmans' Theorem, EOM-CCSD Ionization Energies and Pole Strengths, and Experimental Energies of the Peak Maxima[a]

			K.T.	EOM-CCSD		
band	state	MO	I.E.	I.E.	R_F	exp.[74]
1	^2A_u	$7a_u$	7.73	7.20	0.898	7.64/7.79
	^2A_g	$13a_g$	11.75	7.22	0.875	
2	^2B_g	$6b_g$	8.38	7.44	0.865	7.95/8.15
	^2A_g	$12a_g$	14.01	7.86	0.870	
3	^2A_g	$11a_g$	14.12	8.46	0.868	8.52
4	^2B_g	$5b_g$	14.09	8.81	0.842	9.38
5	^2B_u	$11b_u$	11.81	10.80	0.879	10.36
6	^2A_g	$10a_g$	15.65	12.18	0.857	11.48
7	^2A_u	$6a_u$	14.53	13.12	0.891	12.6
	^2B_u	$10b_u$	14.67	13.27	0.892	
	^2B_g	$4b_g$	14.83	13.35	0.877	
	^2A_g	$9a_g$	16.44	13.45	0.892	
8	^2A_u	$5a_u$	16.35	14.68	0.876	14.2
	^2B_g	$3b_g$	17.31	14.91	0.874	
9	^2B_u			16.01	0.263	15.6
	^2B_u	$9b_u$	17.85	16.08	0.602	
	^2A_u			16.39	0.004	
	^2A_g	$8a_g$	19.05	16.41	0.861	
	^2B_u			16.48	0.007	
	^2B_u			16.98	0.006	
	^2B_g	$2b_g$	24.98	17.29	0.018	
	^2A_u			17.67	0.002	
10	^2A_g	$7a_g$	21.47	18.23	0.034	17.9

[a]All energies are in eV. Experimental results are taken from ref 74.

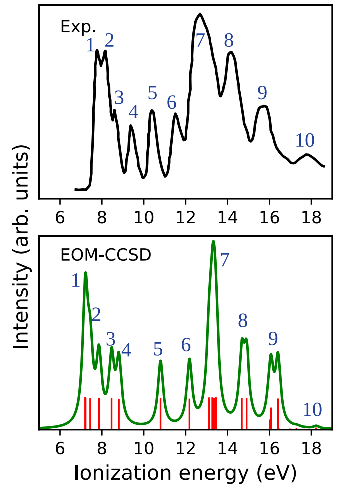

Figure 6. $Ni(C_3H_5)_2$: photoelectron spectrum. The experimental spectrum was redigitized from ref 74.

The lowest IEs are underestimated, and then, progressively overestimated at the high-energy side, probably partly due to the relatively modest basis set employed and incomplete recovery of correlation. Still, differential correlation contributions, of about 5−6 eV for the outer ^2A_g and ^2B_g states, compared to less than 1 eV for the ^2A_u and ^2B_u ones, are completely recovered. It is also interesting to note the strong mixing of HF MOs in the Dyson orbitals. Even if square norms of the Dyson orbitals are still above 0.8, suggesting the validity of the one-electron picture, a large orbital mixing is observed. While the $7a_u^{-1}$ Dyson orbital is practically equal to the $7a_u$ MO (apart from the reduced norm), the (renormalized) composition of the left $13a_g^{-1}$ Dyson is $-0.797{\cdot}13a_g + 0.410{\cdot}9a_g + 0.300{\cdot}10a_g + 0.273{\cdot}12a_g$ MOs; the $12a_g^{-1}$ Dyson is $0.724{\cdot}11a_g - 0.598{\cdot}12a_g + 0.205{\cdot}10a_g + 0.155{\cdot}9a_g + 0.136{\cdot}8a_g$, and so on.

While this "hole mixing" or "orbital rotation" upon ionization has been discussed long ago,[77] it has remained a purely theoretical construct, in the absence of experimental verification. Only transition properties that involve the Dyson orbital can actually probe such an effect, and a prominent one is through photoionization observables. Some representative cross section and asymmetry parameter results for selected ionizations are reported in Figures 7 and 8. A full set is available in Figures S14−S33.

The analysis of our present results, together with previous ones, highlights four important points:

1. Comparison of results relative to HF, DFT, and Dyson orbitals, all with the same DFT continuum, reported in Figure 7, illustrates the importance of correlation in the bound states. The effect is most dramatic in the β parameter. In the case of ionization of the nominal $13a_g$ orbital, HF and Dyson results are very different, illustrating the large reorganization upon ionization. The effect is less marked, but still important, on the cross section. The diminished sensitivity of the cross section is probably due to the fact that it is dominated by the metal 3d component, which remains major in both descriptions. Similar results are obtained also for most (but not all) a_g ionizations (see Figures S14−S20). Analogous considerations hold for the b_g ionizations, see, e.g., $6b_g$ and $5b_g$, which also suggest that upon ionization they mostly exchange role, as HF results for $6b_g$ resemble

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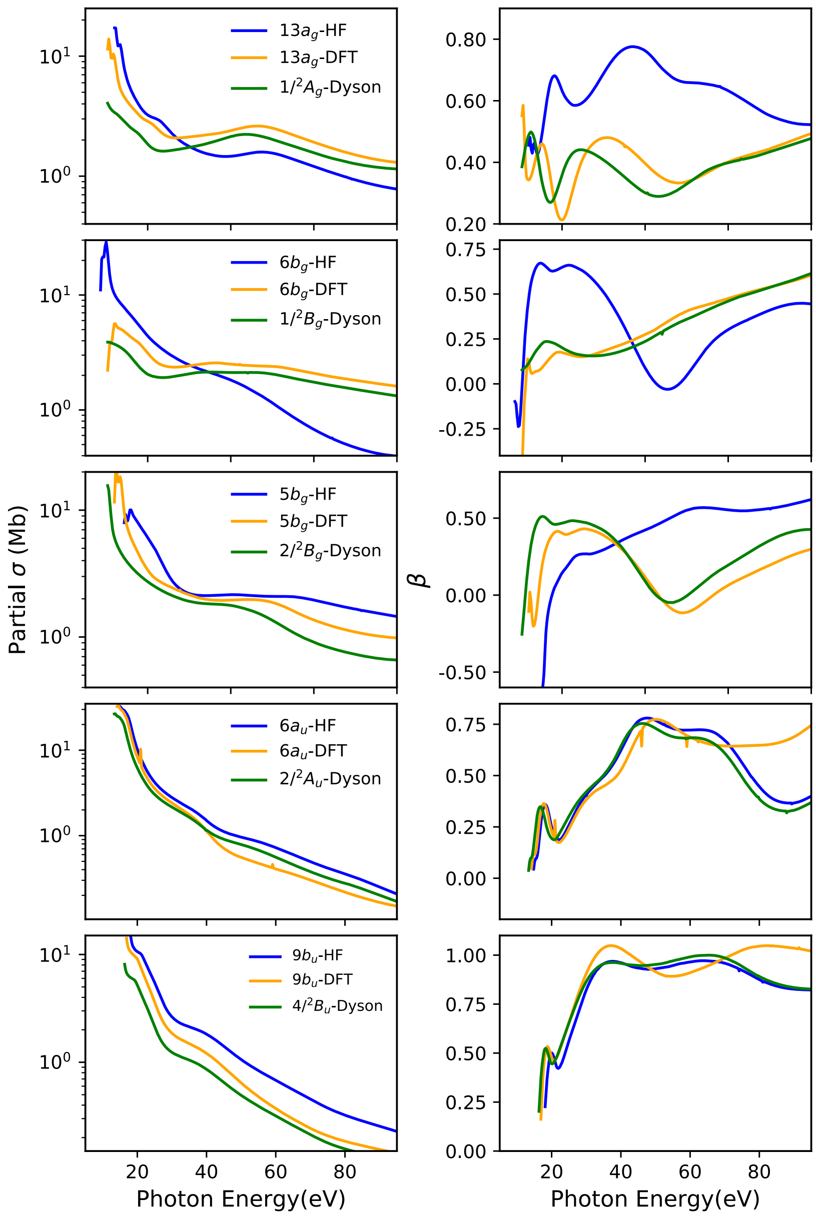

Figure 7. $Ni(C_3H_5)_2$: partial cross sections and asymmetry parameters for particular bound-state ionizations. The B-spline DFT has been used for the description of the continuum orbital.

those of the $2/^2B_g$ Dyson (and $5b_g$ DFT), and vice versa. This is again validated by the fact that the major molecular orbital contributing to $1/^2B_g$ and $2/^2B_g$ Dyson orbitals are indeed $5b_g$ and $6b_g$, respectively. On the other hand, for MOs essentially localized on the organic moiety, the HF and Dyson results are close. We have here chosen $6a_u$ and $9b_u$ for illustration, but the same holds for all a_u and b_u ionizations, that have no metal 3d component, as well as for some inner a_g and b_g ionizations (see Figures S25−S33).

2 Still with reference to Figure 7, it is interesting to compare also the results relative to initial DFT orbitals. As can be seen for $13a_g$, $6b_g$ and $5b_g$, in this case, the results for DFT orbitals are much closer to the Dyson orbitals' ones. It is a clear indication of the superior performance of DFT compared to HF for the description of d orbitals in transition metal complexes, a well-known fact. It appears that the exchange−correlation potential in DFT gives a more balanced description of the interelectronic interaction, while a differential correlation between compact d versus s, p orbitals is much larger in *ab initio*. Curiously, however, DFT performs more poorly for some ligand orbitals, as is clearly seen (especially at higher energies, see Figures S26 and S32) for $6a_u$ and $9b_u$, where HF is closer to Dyson. This behavior offers interesting insights into the nature of DFT orbitals and may suggest specific areas of improvement, complementary to other global properties.

3. The large effect of interchannel coupling has been already highlighted in H_2S and CS_2. Again, this is very specific. In the case of nickel bisallyl, it turns out to be sizable only close to the threshold, and, depending on the orbital, of modest or minor importance, as can be seen for the same orbitals in Figure 8. But the striking effect is the appearance of autoionization resonances in the region of metal 3p → 3d excitation, at about 70 eV. This is a very

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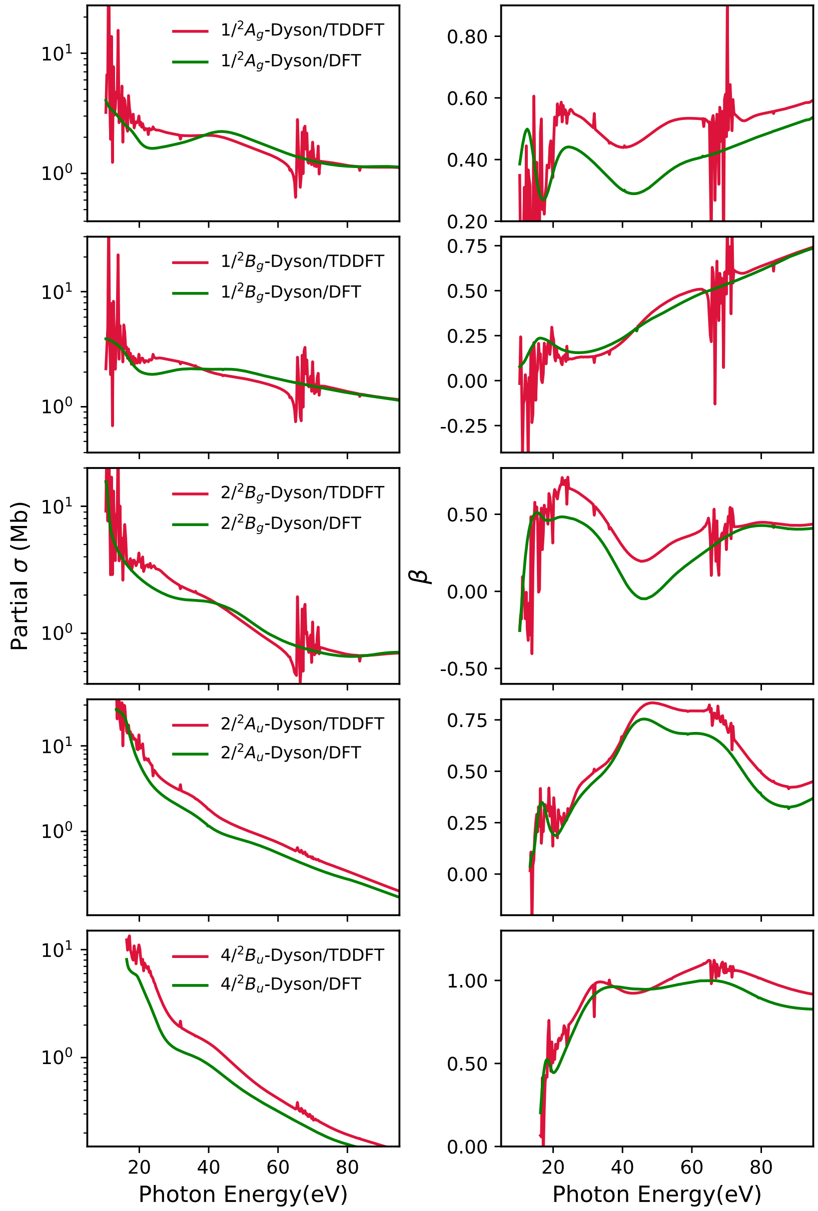

Figure 8. $Ni(C_3H_5)_2$: partial cross section and asymmetry parameters for particular bound-state ionizations. EOM-CCSD Dyson orbitals have been used for the description of the bound part.

strong signature of the metal d participation in the orbital ionized, and, in fact, it is strongly present in the a_g and b_g ionizations relative to Ni 3d orbitals, and are instead barely visible in the a_u and b_u ligand bands, which can have no metal 3d content, and appear only via weak continuum−continuum coupling with the other channels.

4. Clearly, photoionization observables are a powerful tool for spectral assignment. In fact, this has been used since the inception of photoelectron spectroscopy as a change in band intensities in going from HeI to HeII radiation, and to X-rays, or as differences in the β behavior between σ and π orbitals, but it has become much more informative with the advent of tunable synchrotron radiation. It is well known that upon comparing ionization cross-section profiles from 2p orbitals of the second-row elements with 3d orbitals of transition metals, the former displays a rapidly decaying cross section with a higher value near the threshold. Based on this, we can comment from Figure 9 that the outermost bands are due to ionizations from the 3d orbitals of the Ni atom, except that corresponding to the $7a_u$ orbital. This Dyson orbital is similar to the π orbital on the C_3H_5 moiety. On the other hand, all of the ionization pathways constituting bands 5−10 mimic the profile for the 2p orbital of the C atom. Thus, these are of ligand origin.

As a single example, we report, in Figure 10, the ratio of the $7a_u$ to $13a_g$ cross sections, which are expected to be the two outermost ionizations. A careful high-resolution experiment could probably assess which is the HOMO band if they are not practically degenerate, and in general to disentangle the assignment of the ionizations composed of in bands 1−3. A more detailed analysis based on pure DFT/TDDFT results was presented in a previous study.[78]

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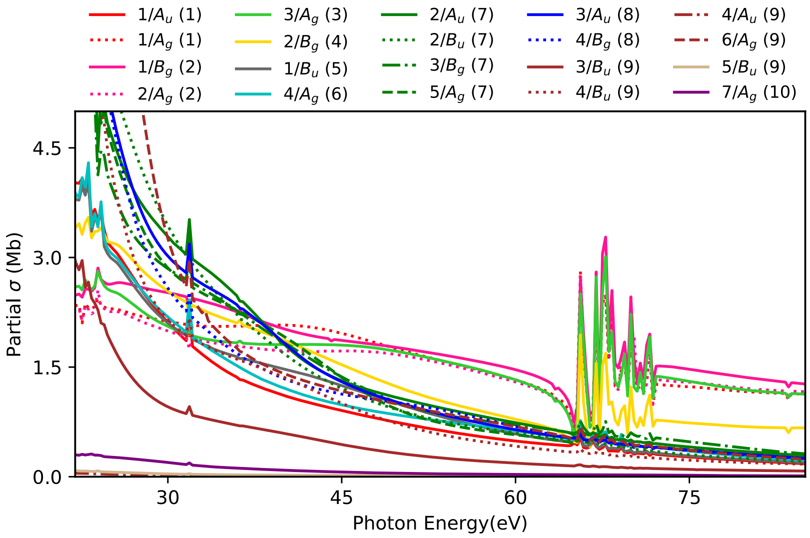

Figure 9. $Ni(C_3H_5)_2$: partial cross section of the ionizations for bands 1−10, computed using the EOM-CCSD Dyson/TDDFT approach. The band to which the ionization belongs is given in parentheses.

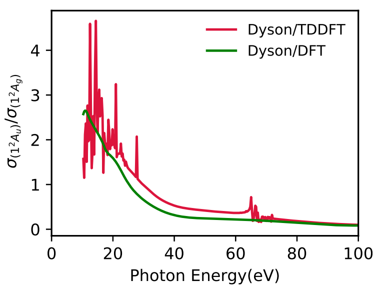

Figure 10. $Ni(C_3H_5)_2$: ratio of a partial cross section of the ionizations from the two outermost bound channels.

5. CONCLUSIONS

In summary, our devised methodology, coupling EOM-CCSD Dyson orbital coefficients to describe the bound-state character of the target system and B-spline TDDFT continuum orbitals, gives a precise depiction of the photoionization properties. The developed method acts as a more accurate extension of the EOM-CCSD Dyson/KS-DFT approach described earlier.[16] However, this extra noniterative step over KS-DFT is generally necessary only for reproducing highly sophisticated experimental observations. We observe that for the case of ionizations from the outer-valence orbitals originating from 2s, 2p orbitals of second-row atoms, only a slight improvement is seen upon using TDDFT continuum over DFT continuum orbitals. Although, as all of the corresponding Dyson orbitals are practically equivalent to the molecular orbital considered for Hartree−Fock or DFT calculation, no drastic improvement is noticed. However, the formalism is an absolute necessity in the case of treatment of interchannel couplings, the precise positioning of Cooper minima, and to account for correlated ionization states, like satellite states, which are absent in Koopmans' treatment. Moreover, our approach accounts for auto-ionization resonances lacking in its DFT equivalent. We also manifest that in the case of transition metal complexes, Koopmans' theorem often fails due to large correlation and relaxation effects, which also cause dramatic changes of Dyson orbitals with respect to *ab initio* ones. Interestingly, in such cases, DFT orbitals are much closer to Dyson than to HF ones. The opposite is found, however, for other ionizations. The interpretation of photoelectron data is then possible only upon using a highly correlated bound target state description, as exemplified in the case of nickel bisallyl complex.

The approach presented here is completely general and computationally efficient. It allows for the calculation of all photoionization dynamical variables. We here present results for partial cross sections, branching ratios, asymmetry parameters, and molecular frame photoelectron angular distributions. It is seen that the asymmetry parameter is the one most significantly improved by our current approach. Based on the molecules studied, we also conclude that generally molecules with third (or higher)-period elements, are better described by the EOM-CCSD Dyson/TDDFT approach. The appearance of the Cooper minima in the asymmetry parameters is very characteristic of third-row atoms (here, S 3p orbital).

Furthermore, theoretical advancements are possible building on this formalism, like the inclusion of higher-order correlation effects in the treatment of the bound states. Open-shell systems can, in principle, also be treated within the proposed formalism and will be the subject of future studies. The method is also suitable for extension to encompass nuclear dynamic problems to describe the vibrational resolution of the experimental spectra. Another interesting application can be for the case of time-resolved experiments, as the photoionization process is highly sensitive to the bound-state nature of the ionizing orbital.

Finally, although we have not considered satellite states in detail in this work, photoionization observables are a powerful signature of such states, giving information on the nature of their electronic configuration and parentage of primary states, if any. Since they are strictly forbidden at the one-particle level, they are a direct probe of correlation effects and a challenging task for highly correlated theoretical approaches. In fact, while originally raising great interest, the inadequacy of the theoretical tools of the time and experimental difficulty in accurately measuring such often weak and poorly resolved features, caused decline in interest. Present facilities are now at a point that a renewed attack on the problem, both experimentally and theoretically, is warranted. Since the EOM-CCSD approach employed here is still only partly adequate, we have considered only one well-characterized satellite in CS_2, where already a rather good reproduction of the experimental data is afforded. More detailed

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studies, including branching ratios with primary states, of the several satellites that are detected in all these molecules, may give a more precise characterization of the nature of the multielectron excitations and the improvement of the theoretical tools needed for a quantitative description. Among these improvements is a more accurate account of double-excitation effects, as our preliminary results using a perturbatively corrected ionization energy of the satellite state of CS_2 indicate. Work in this direction is in progress.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.1c00303.

> Computed photoelectron spectra versus experiment; comparison of R_F computed using different definitions; characterizing the satellite band of CS_2; MFPADs of CS_2; results and discussion for C_3O_2 and C_3S_2; for $Ni(C_3H_5)_2$: EOM-CCSD/aug-cc-pVDZ PES and individual target bound-state partial cross section and asymmetry parameters; geometries (Cartesian coordinates) of all molecules studied (PDF)

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Notes

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